

Week 7 logistic regression and classification

1. What is the sigmoid function? What does it map the input to? Is it monotone? What do we use it to in the logistic function?
2. What is the derivative to the sigmoid function?
3. What is logistic regression? Is logistic regression a classifier or regression?
4. What are the steps to predict in logistic regression? What is the typical threshold?
5. How do we calculate the probability of $\mathbf{x}$ belonging to class 1 $P(t=1|\mathbf{x})$? how do we calculate $P(t=0|\mathbf{x})$
6. What is the goal of training?
7. What is cross entropy loss and how do we calculate it?
8. What is the chain rule for derivatives?
9. How do we find the derivative in logistic regression?
10. What is the difference in using cross-entropy loss or MSE when calculating loss in logistic regression?
11. What is batch training and stochastic gradient descent? Also what is mini-batch training?
12. What is one-hot encoding?
13. What is one-vs-rest
14. What is multi-label perceptron?
15. What is multi-class perceptron?
16. What is multi-output linear regression?
17. What is Multinomial logistic regression